POWER BI REPORT GUIDE

Agricultural production and food security monitor

How the FAO report was built, what each page shows, how to use the dashboards, and how to maintain and extend the project.



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Report author · StarTechOne Nigeria Limited

Report: FAO project.pbip (17 pages: 13 story pages, 3 dashboards, 1 drill-through)
Data: Project_07_Agriculture.xlsx: 25 countries, 11 crops, 17 survey years (1990-2022)
Tools: Power BI Desktop (PBIP format), Power BI modeling MCP, powerbi-report-author CLI, Node.js
Version: October 2026

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> **Who this guide is for.** Policy and investment readers who will read or present the report (sections 1, 2, 7, 8 and 9), and analysts who will maintain or extend it (sections 3 to 6 and 10 to 13).

1 · EXECUTIVE SUMMARY

Output grew 65% since 1990, but yield has not moved since 2014

29.04bn t
production, all 17 survey years

4.54 t/ha
average yield

18 of 25
countries below their regional benchmark

5.12bn t
extra output if those gaps closed (+17.6%)

Verdict	Topic	Evidence
STALLED	Production growth	Up 61.5% from 1990 to 2014 (1.22bn to 1.98bn t a year), then only 2.3% more by 2022 (2.02bn t)
FLAT	Yield	3.60 t/ha in 1990, 4.98 in 2014, still 4.98 in 2022. Growth to 2014 came from yield (+38%) more than land (+17%)
GAP	Africa	Grows 42.8% of all output at the lowest regional yield, 3.67 t/ha (Americas 6.85)
STRENGTH	Roots and tubers	Use 14.9% of the farmland but grow 43.9% of the tonnage. Cereals use 63.3% of the land for 50.3%
OPPORTUNITY	Closing yield gaps	Six countries hold over half of the 5.12bn t potential: Philippines, Argentina, Brazil, Malawi, Uganda, India

Recommended actions

1. **Close Africa's yield gap.** Seeds, fertiliser and extension in the 14 African countries, where yields trail the regional benchmark of 4.23 t/ha. Malawi and Uganda alone could add about 0.8bn t.
2. **Restart yield growth.** Put research and irrigation money into the six largest-gain countries. Target: yield growth back above 2% per survey round.
3. **Back roots and tubers.** Fund storage and processing for cassava and yams so the harvest reaches markets. Target: keep their share of output above 40%.

Find where higher yields would add the most food

The project was briefed as an agricultural production and food security monitor for policy and investment decision-makers. The report answers five questions:

1. Is production still growing, and is the growth coming from yield or from more land?
2. Which regions and countries produce the most, and at what yield?
3. Which crops give the most food per hectare?
4. How far is each country below its regional top-quartile yield?
5. Where would closing that gap add the most tonnes?

> **Scope note.** The brief refers to the Sahel, but the data covers 25 countries on four continents (Africa 14, Americas 4, SE Asia 4, W Pacific 3). The report describes the data as it is and does not claim Sahel-specific findings.

3 · THE SOURCE DATA

One workbook: a production fact table and four lookup sheets

Source: `C:\Users\USER\Desktop\project\Project_07_Agriculture.xlsx`. The model refreshes from this path.

Sheet	Grain	Key columns
FactProduction	Country × crop × survey year × element	CountryCode, CropID, YearKey, ElementID, Value, Unit, IsEstimate
DimCountry	25 countries	CountryName, ISO3, FAORegion, IncomeGroup, IsLDC, Population_M
DimCrop	11 crops (7 staples)	CropName, CropFamily (Cereals, Roots & Tubers, Pulses, Cash Crops), CaloriesPerKg, IsStaple
DimYear	17 survey years	Year (1990, 1992 … 2022), Decade
DimElement	Measure type	PR production (t), AH area harvested (ha), YD yield, imports, exports

Data quality

- 252 values are blank (78 area, 84 production, 90 yield). They are left out of the sums.
- 14.6% of records are flagged as FAO estimates rather than official figures.
- Population figures are not realistic (China 13M, India 164M), so per-person measures are not used.
- No duplicate or orphan keys were found.

A star schema: one fact table, four dimensions and a measures table

From	To	Cardinality
FactProduction[CountryCode]	DimCountry[CountryCode]	Many to one, single direction
FactProduction[CropID]	DimCrop[CropID]	Many to one, single direction
FactProduction[YearKey]	DimYear[YearKey]	Many to one, single direction
FactProduction[ElementID]	DimElement[ElementID]	Many to one, single direction

All measures sit in the `DAX` table. Measures added for this version are in the display folder **Report v2**.

The measures the report uses

Measure	Definition	Format
Total Production	Sum of Value where element = Production (tonnes)	#,0
Total Area Harvested	Sum of Value where element = Area Harvested (ha)	#,0
Average Yield	Total Production / Total Area Harvested (t/ha)	0.00
Regional Top-Quartile Yield	75th percentile (PERCENTILEX.INC) of country yields within the region(s) in context	0.00
Yield Gap	MAX(benchmark − yield, 0)	0.00
Potential Production Gain	Sum over countries of yield gap × area harvested (tonnes)	#,0
Potential Gain %	Potential gain / Total Production	0.0%
Gap Countries	Countries with a yield gap above zero	0
Production Share %, Area Share %	Share of the all-country, all-crop total in the current year selection	0.0%
Survey Years, Production per Survey Year	Distinct years with production data; production / survey years	0, #,0
Production 1990 / 2014 / 2022, Yield 1990 / 2014 / 2022	The measure fixed to one survey year	#,0, 0.00
Countries Covered, Crop Count, Staple Crops	Distinct counts	0
Top Producing Crop, Top Crop Family	Name of the largest item by production	text
Estimated Records %	Share of fact rows flagged IsEstimate	0.0%

```
Regional Top-Quartile Yield =
VAR _currentRegions = VALUES ( DimCountry[FAORegion] )
VAR _yieldTable =
    ADDCOLUMNS (
        FILTER ( ALL ( DimCountry ), DimCountry[FAORegion] IN _currentRegions ),
        "@CountryYield", CALCULATE ( [Average Yield] ) )
RETURN
    PERCENTILEX.INC ( FILTER ( _yieldTable, NOT ISBLANK ( [@CountryYield] ) ), [@CountryYield], 0.75 )

Potential Production Gain =
SUMX ( VALUES ( DimCountry[CountryCode] ),
       VAR _gap = [Yield Gap] RETURN _gap * [Total Area Harvested] )
```

6 · CORRECTIONS

Six things fixed from the original report

#	Issue	Fix
1	The old insights page had the region shares swapped (Americas 42.76%, Africa 18.58%)	Africa is 42.8%, Americas 24.7%, SE Asia 18.6%, W Pacific 13.9%
2	"Top-quartile" benchmark averaged the top three-quarters of countries	Now the true 75th percentile. Africa's benchmark moves from 3.90 to 4.23 t/ha
3	Potential gain divided by 10,000 for an hg/ha conversion, but yield is already t/ha	Division removed: the gain is 5.12bn t, not a tiny number
4	Totals add up every survey year without saying so	Labels say "all survey years"; trend pages show one year at a time
5	Calories per person used unrealistic population figures	Per-person measures are not shown
6	Brief described the Sahel	Report describes the actual four-region scope

7 · REPORT WALKTHROUGH

Story pages: thirteen pages, one finding each

Each story page has a Georgia title that states the finding, grey context bars with one highlighted bar, a source line and a Dashboards button.

1 Cover

A rising sun over ploughed fields and wheat (drawn by _build\\make-cover.ps1), with the title, the author block (photo, name, StarTechOne Nigeria Limited) and a harvest-gold start button. **Start the story** opens page 2.



2 Brief

The aim, the five questions, and scope tiles: 25 countries, 11 crops, 17 survey years, 7 staples. Notes that the data is wider than the Sahel.

The brief: find where higher yields would add the most food

Aim
Show decision-makers how staple-crop production and yield have moved since 1990, where yields trail the best in each region, and where closing that gap would add the most food.

Questions this report answers

- Is production still growing, and is the growth coming from yield or from more land?
- Which regions and countries produce the most, and at what yield?
- Which crops give the most food per hectare?
- How far is each country below its regional top-quartile yield?
- Where would closing that gap add the most tonnes?

25 countries	**11** crops
17 survey years, 1990-2022	**7** staple crops

Four FAO regions: Africa (14 countries), Americas (4), SE Asia (4), W Pacific (3).
The brief mentions the Sahel; the data covers four continents, so the report describes it as it is.

Source: Project_07_Agriculture.xlsx · scope of the data · 2

Dashboards

3 Scorecard

Four headline KPIs and five findings, each with a verdict pill: STALLED, FLAT and GAP in terracotta; STRENGTH and OPPORTUNITY in green.

Output grew 65% since 1990, but yield has not moved since 2014

29.04bn tonnes produced, all survey years	**6.40bn** hectares harvested	**4.54** average yield, t/ha	**5.12bn** extra tonnes if yield gaps closed

STALLED	**Production growth**	Up 61.5% from 1990 to 2014, then only 2.3% more by 2022
FLAT	**Yield**	3.60 t/ha in 1990, 4.98 t/ha in 2014, still 4.98 t/ha in 2022
GAP	**Africa**	Grows 42.8% of all output at the lowest regional yield, 3.67 t/ha
STRENGTH	**Roots and tubers**	Use 14.9% of the farmland but grow 43.9% of the tonnage
OPPORTUNITY	**Closing yield gaps**	18 of 25 countries trail their benchmark; closing it adds 5.1bn t (17.6%)

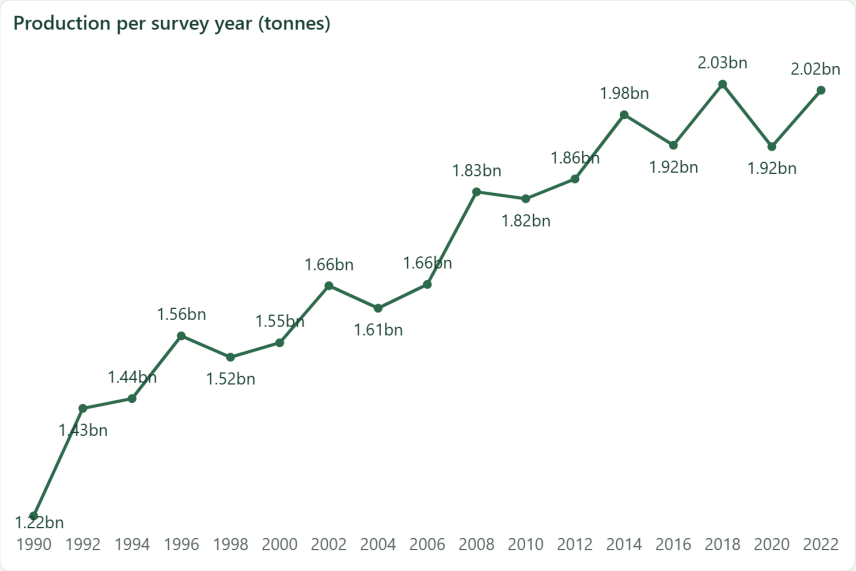
Source: Project_07_Agriculture.xlsx · totals are cumulative over the 17 survey years · 3

Dashboards

4 Production trend

Production per survey year: 1.22bn t in 1990, 1.98bn in 2014, 2.02bn in 2022. Growth to 2014 came from yield more than land.

Production rose 61% from 1990 to 2014, then grew only 2% in the next eight years



1.22bn
tonnes in 1990

1.98bn
tonnes in 2014 (+61.5%)

2.02bn
tonnes in 2022 (+2.3% since 2014)

Growth came from yield, not land.
Area rose 17% to 2014 while yield rose 38%. After 2014 both grew about 2%.

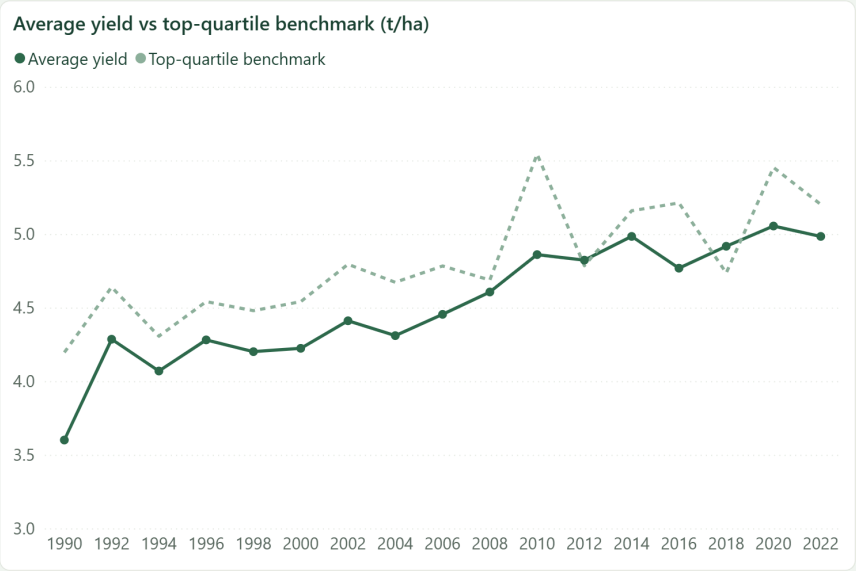
Source: Project_07_Agriculture.xlsx · one value per survey year · 4

Dashboards

5 Yield trend

Average yield against the top-quartile benchmark: up from 3.60 to 4.98 t/ha by 2014, flat since.

Yield climbed from 3.60 to 4.98 t/ha by 2014 and has not moved since



3.60
t/ha in 1990

4.98
t/ha in 2014 (+38%)

4.98
t/ha in 2022 (no change)

The benchmark is the 75th percentile of country yields.
A country at the benchmark out-yields three in four of its peers.

Source: Project_07_Agriculture.xlsx · yield = production / area harvested, per survey year · 5

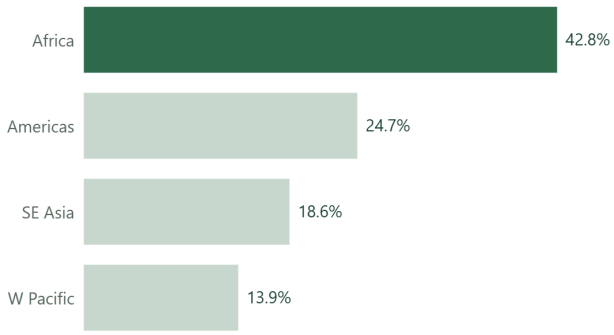
Dashboards

6 Regions

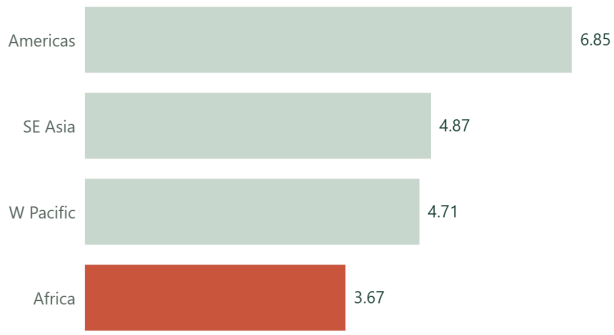
Africa grows 42.8% of output (green) at the lowest yield, 3.67 t/ha (terracotta).

Africa grows 43% of the output at the lowest yield of any region, 3.67 t/ha

Share of production by region



Average yield by region (t/ha)



Africa farms the most land at the lowest yield. Raising it to the Americas level would almost double its output.

Source: Project_07_Agriculture.xlsx · totals are cumulative over the 17 survey years · 6

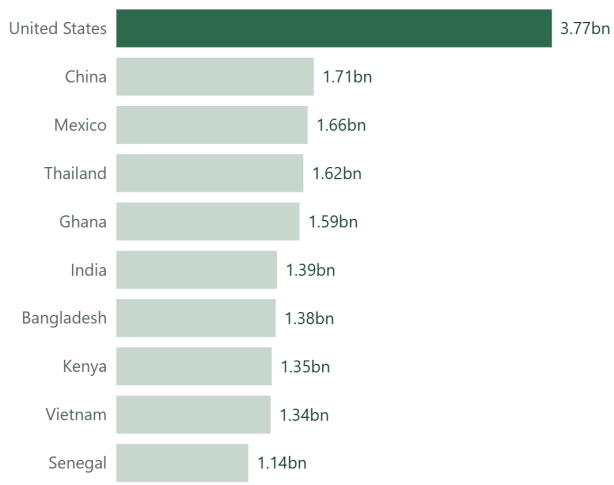
Dashboards

7 Top producers

Top 10 producers, with the United States highlighted at 13% of output, and a top-15 table with share and yield.

Five countries grow 36% of all output; the United States alone grows 13%

Top 10 producers (tonnes)



Top 15 countries: production, share and yield

Country	Production (t)	Share	Yield (t/ha)
United States	3,772,522,340	13.0%	10.57
China	1,710,087,526	5.9%	5.89
Mexico	1,657,028,455	5.7%	5.68
Thailand	1,619,121,835	5.6%	7.50
Ghana	1,586,367,213	5.5%	4.15
India	1,390,597,988	4.8%	4.13
Bangladesh	1,380,456,940	4.8%	4.49
Kenya	1,345,703,265	4.6%	4.60
Vietnam	1,335,732,349	4.6%	4.80
Senegal	1,142,952,092	3.9%	4.46
Egypt	1,124,007,762	3.9%	3.49
Indonesia	1,005,669,893	3.5%	4.05
Philippines	1,005,371,963	3.5%	3.45
Brazil	929,106,112	3.2%	4.63
Ethiopia	928,737,646	3.2%	3.53

Source: Project_07_Agriculture.xlsx · totals are cumulative over the 17 survey years · 7

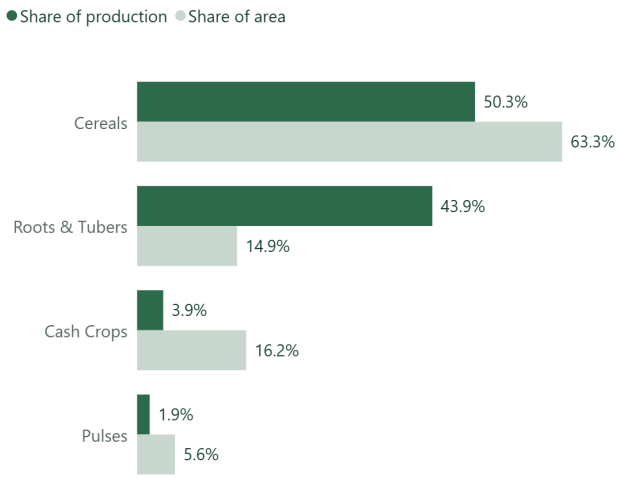
Dashboards

8 Crop families

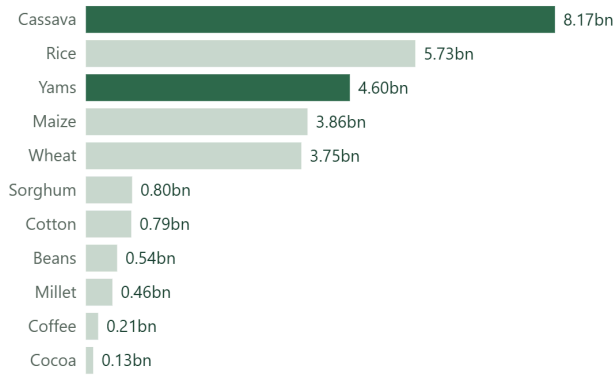
Share of production against share of area: roots and tubers grow 43.9% of output on 14.9% of land. Includes a calorie caveat.

Roots and tubers use 15% of the farmland but grow 44% of the tonnage

Share of production vs share of area



Production by crop (tonnes)



Tonnes are not calories: cassava carries about 1,100 kcal/kg, maize 3,700.

Source: Project_07_Agriculture.xlsx · totals are cumulative over the 17 survey years · 8

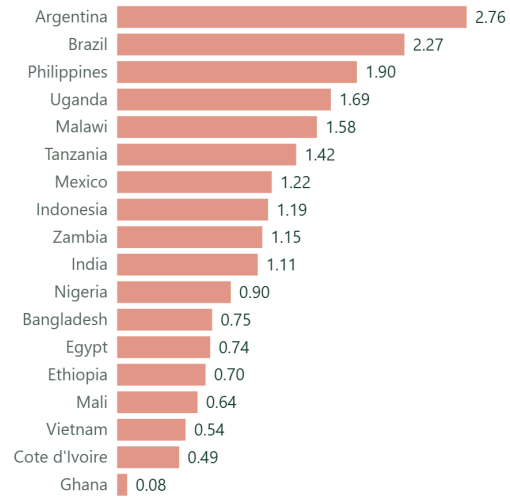
Dashboards

9 Yield gap

The 18 countries below their regional benchmark, largest gap first, with a table of yield, benchmark and gap.

18 of 25 countries yield below the top quartile of their own region

Yield gap to regional benchmark (t/ha)



15 largest gaps: yield vs regional benchmark (t/ha)

Country	Region	Yield	Benchmark	Gap
Argentina	Americas	4.14	6.90	2.76
Brazil	Americas	4.63	6.90	2.27
Philippines	W Pacific	3.45	5.34	1.90
Uganda	Africa	2.54	4.23	1.69
Malawi	Africa	2.65	4.23	1.58
Tanzania	Africa	2.81	4.23	1.42
Mexico	Americas	5.68	6.90	1.22
Indonesia	SE Asia	4.05	5.25	1.19
Zambia	Africa	3.08	4.23	1.15
India	SE Asia	4.13	5.25	1.11
Nigeria	Africa	3.33	4.23	0.90
Bangladesh	SE Asia	4.49	5.25	0.75
Egypt	Africa	3.49	4.23	0.74
Ethiopia	Africa	3.53	4.23	0.70
Mali	Africa	3.59	4.23	0.64

Source: Project_07_Agriculture.xlsx · benchmark = 75th percentile of country yields in the region · 9

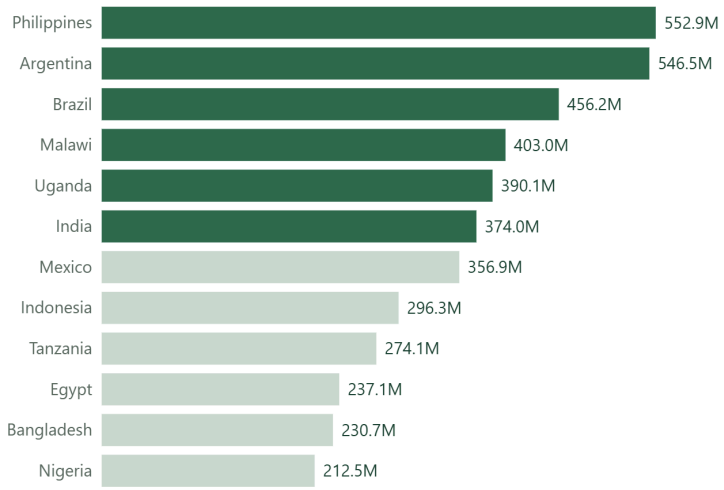
Dashboards

10 Potential gain

Extra tonnes per country at benchmark yield: 5.12bn t in total, over half of it in six countries.

Closing the gap would add 5.1bn t; six countries hold over half of it

Extra tonnes if each country reached its benchmark



5.12bn
extra tonnes, all survey years

17.6%
on top of actual production

18
countries below benchmark

Gain = yield gap x area harvested.
Large farm areas, not only large gaps, put Argentina and Brazil near the top.

Source: Project_07_Agriculture.xlsx · totals are cumulative over the 17 survey years · 10

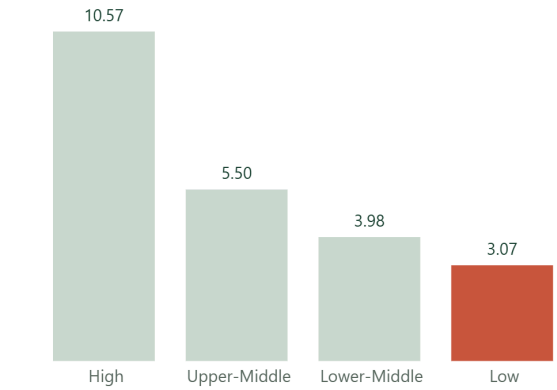
Dashboards

11 Income and yield

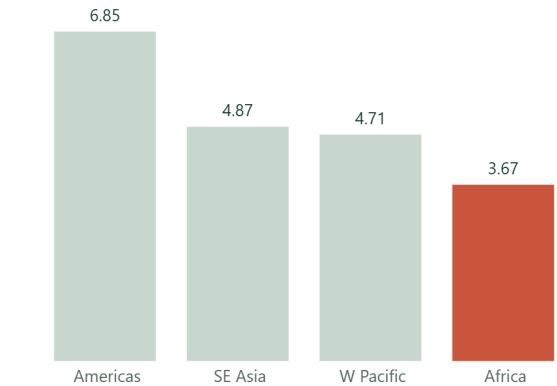
Yield by income group and by region, with a note on small groups.

High-income farms yield 10.6 t/ha, more than three times low-income farms at 3.1

Average yield by income group (t/ha)



Average yield by region (t/ha)



Read with care: the High group is one country (the United States), and all four Low-income countries are in Africa.

Source: Project_07_Agriculture.xlsx · totals are cumulative over the 17 survey years · 11

Dashboards

12 Recommendations

Three actions with targets, backed by the gain, gap-country and gain % KPIs.

Three actions: close Africa's gap, restart yield growth, back roots and tubers

1

Close Africa's yield gap

Seeds, fertiliser and extension in the 14 African countries, where yields trail the regional benchmark of 4.23 t/ha.

Target: Malawi and Uganda alone could add about 0.8bn t

2

Restart yield growth

Yield has been flat at 4.98 t/ha since 2014. Put research and irrigation money into the six largest-gain countries.

Target: yield growth back above 2% per survey round

3

Back roots and tubers

Cassava and yams give the most tonnes per hectare. Fund storage and processing so the harvest reaches markets.

Target: keep their share of output above 40%

5.12bn
extra tonnes at benchmark yield

18
countries below benchmark

17.6%
more food from the same land

Source: Project_07_Agriculture.xlsx · targets are planning goals, not forecasts · 12

Dashboards

13 Data notes

How to read totals, the corrections, data limits and the estimate share by region.

Assumptions and data issues behind these numbers

How to read the totals

- Totals add up all 17 survey years (1990-2022, every second year). Trend pages show one year at a time.

Corrections to the original report

- Benchmark: the old measure averaged the top three-quarters of countries; it is now the 75th percentile.
- Potential gain: a divide-by-10,000 hg/ha conversion was removed; yield is already in t/ha.
- Region shares were swapped on the old insights page; Africa is 42.8%, not the Americas.

Data limits

- Population figures are not realistic (China 13M), so per-person measures are not used.
- 252 values are blank (78 area, 84 production, 90 yield) and are left out of the sums.

14.6%
of records are FAO estimates, not official figures

Estimates and coverage by region

Region	Estimated	Countries	Yield (t/ha)
Africa	14.6%	14	3.67
Americas	14.1%	4	6.85
SE Asia	14.5%	4	4.87
W Pacific	15.2%	3	4.71

Estimates are spread evenly.
Every region sits between 14.1% and 15.2%, so they do not tilt the regional comparisons.

Source: Project_07_Agriculture.xlsx · all records · 13

Dashboards

Dashboards: three interactive pages and a country profile

The dashboards use a bento layout. A dark hero tile carries the headline number, a 2 x 2 cluster holds the supporting KPIs, and a wide trend tile sits beside them. Below are tiles of mixed widths with a tinted "what this shows" tile. Survey year, Region and Crop filters stay in sync across all three. The tabs switch dashboards or return to the story, and the author block sits bottom-right.

D1 Production overview

Dark hero tile with total production; a 2 x 2 cluster with area, yield, countries and survey years; the production trend; production by region; top 8 producers; and a tinted "what this shows" tile.



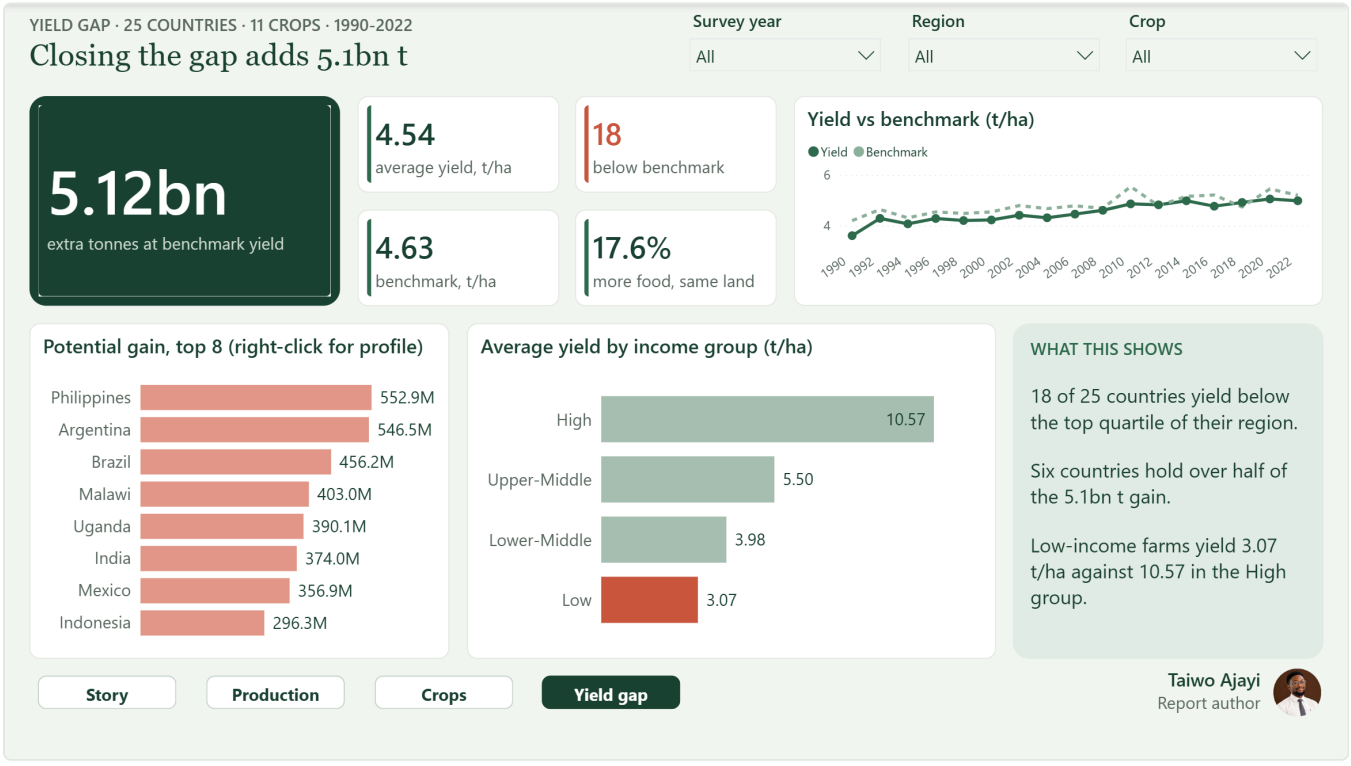
D2 Crop analysis

Hero tile names the largest crop. The KPI cluster shows total production, the largest crop family, crop count and staples. Tiles below show crop family share of production against area, production by crop, and yield by crop.



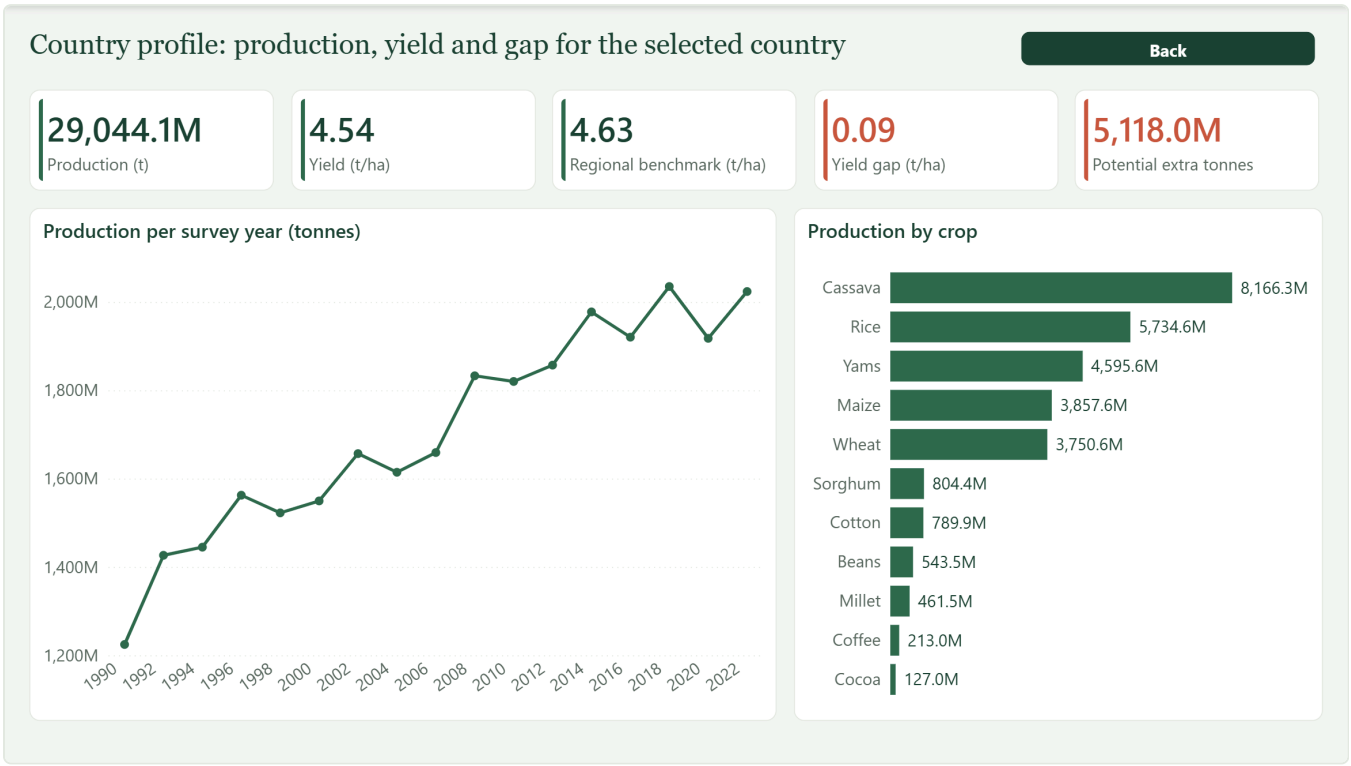
D3 Yield and gap

Hero tile shows the potential gain. The KPI cluster shows yield, benchmark, gap countries and gain %. Tiles below show yield against benchmark over time, the top 8 countries by gain, and yield by income group.



Country profile

Hidden drill-through page. Right-click a country bar and choose Drill through, then Country profile, to see that country's KPIs, its trend and its crop mix. **Back** returns you to the page you came from.



Reading, filtering, drilling and presenting

Navigating

- Start on the cover and press **Start the story**, or use the page tabs. In Desktop, hold Ctrl when clicking a button.
- Every story page has a **Dashboards** button; the dashboards have a **Story** tab back.

Filtering

- Pick one or more survey years to see a single year's totals instead of the cumulative 1990-2022 figure.
- Region and Crop filters narrow every visual on the page and carry over to the other dashboards.
- Clicking a bar cross-filters the other visuals on the page; click it again to clear.

Drilling to a country

- Right-click any country bar (D1 top producers, D3 potential gain) and choose **Drill through → Country profile**.
- The profile shows that country's production, yield, benchmark, gap, potential gain, trend and crop mix. Press **Back** to return.

Presenting

- Use the story pages in order for a briefing; switch to the dashboards for questions.
- Quote cumulative totals as "1990-2022, all survey years"; quote one-year figures from pages 4 and 5.

Calm and editorial, recoloured for agriculture

Forest #1B4332
titles, text, cover

Green #2D6A4F
accent, highlight

Sage #CBD8CF
context bars

Terracotta #C8553D
problems only

- Page #F3F5F2; white tiles with a #E2E8E0 hairline and 10px radius.
- Georgia for titles; Segoe UI for everything else; nothing under 12pt.
- KPI tiles carry a 4px left accent rule: green for normal, terracotta for problem measures.
- One highlighted bar per chart; the rest stay sage so the eye goes to the point.
- Tables shade the key column with a soft gradient instead of heavy colour.
- Quiet tab navigation; the author's name and photo on the cover and every dashboard.
- The same design system as the Bank Churn report, built from one shared generator.

Tie-out values (no filters, all survey years unless stated)

Check	Value
Total production / area / yield	29.04bn t / 6.40bn ha / 4.54 t/ha
Production 1990 / 2014 / 2022	1.22bn / 1.98bn / 2.02bn t
Yield 1990 / 2014 / 2022	3.60 / 4.98 / 4.98 t/ha
Region share: Africa / Americas / SE Asia / W Pacific	42.8% / 24.7% / 18.6% / 13.9%
Region yield: Africa / Americas / SE Asia / W Pacific	3.67 / 6.85 / 4.87 / 4.71
Regional benchmark: Africa / Americas / SE Asia / W Pacific	4.23 / 6.90 / 5.25 / 5.34
Crop family share of production / area: Cereals	50.3% / 63.3%
Roots & Tubers	43.9% / 14.9%
Top crops: Cassava / Rice / Yams / Maize / Wheat	8.17 / 5.73 / 4.60 / 3.86 / 3.75bn t
Top producers: United States / China / Mexico / Thailand / Ghana	3.77 / 1.71 / 1.66 / 1.62 / 1.59bn t
Yield by income: High / Upper-Middle / Lower-Middle / Low	10.57 / 5.50 / 3.98 / 3.07
Potential gain / gap countries	5.12bn t / 18
Largest gains: Philippines / Argentina / Brazil / Malawi / Uganda / India	553 / 547 / 456 / 403 / 390 / 374 Mt
Estimated records	14.6%

How to refresh, change and rebuild the report

Refresh the data

Replace `Project_07_Agriculture.xlsx` at the same path with the same sheet and column names, open `FAO project.pbip` and press **Refresh**. Then update the numbers in titles and notes, which are written text.

Change a page

Pages are generated. Edit `_build\generate-report.js`, then run:

```
cd "C:\Users\USER\Desktop\ALL PROJECTS\FAO project"
node _build\generate-report.js
powerbi-report-author validate "FAO project.Report"
```

Close Power BI first (or apply external changes) so Desktop does not save its older copy over the new pages. Shared building blocks live in `ALL PROJECTS\_shared\pbir-kit.js` and are used by every project.

Change a measure

Edit it in Power BI Desktop and save, or edit `FAO project.SemanticModel\definition\tables\DAX.tmdl` with Desktop closed. `_build\fix-model.js` re-applies this version's measure fixes and is safe to run twice.

What the numbers can and cannot tell you

- Totals are cumulative over the 17 survey years; divide by 17 (or pick one year) for an annual figure.
- Tonnes are not food value: cassava and yams carry about 1,100 kcal/kg against 3,700 for maize, so tonnage flatters roots and tubers.
- The benchmark compares each country only with others in its own region; regions with few countries (W Pacific has 3) give a less stable benchmark.
- Potential gain assumes the same harvested area at benchmark yield; it is a ceiling, not a forecast.
- The High income group is a single country (the United States); all four Low-income countries are in Africa.
- 14.6% of records are estimates and 252 values are blank.
- Survey years are every second year, so year-on-year measures compare two-year steps.

APPENDIX A

Project files

Path (in ALL PROJECTS\FAO project)	What it is
`FAO project.pbip`	Open this in Power BI Desktop
`FAO project.SemanticModel\`	The data model as TMDL text files
`FAO project.Report\`	The report as PBIR files. Generated; do not edit by hand
`_brief\report-spec.md`	Approved report spec
`_build\generate-report.js`	Generator for the theme and all report pages
`_build\fix-model.js`	Re-applies the measure fixes and additions to the model files
`_build\assets\taiwo-ajayi.png`	Author photo used on the cover and dashboards
`_build\backup\`	The original report before the redesign
`power bi project Ajayi Taiwo Anthony.pdf`	Export of the original report
`FAO_Report_Guide.pdf`	This guide